Quaternion Attitude Dynamics, Nonlinear PD/PID Attitude Control & Smooth Trajectory Tracking for a Rigid Body and Missile Pitch Channel

1 Attitude Representation with Unit Quaternions

1.1 Axis–Angle to Quaternion

The attitude is stored as a scalar-first unit quaternion $\boldsymbol{q} = [q_0,\ q_1,\ q_2,\ q_3]^T = [q_0,\ \boldsymbol{q}_v^T]^T$. For a rotation of angle ϕ about an axis $\boldsymbol{e}$ (normalised to unit length, $\hat{\boldsymbol{e}} = \boldsymbol{e}/\|\boldsymbol{e}\|$):

$$\boldsymbol{q} = \begin{bmatrix} \cos(\phi/2) \\ \hat{\boldsymbol{e}}\,\sin(\phi/2) \end{bmatrix}, \qquad \boldsymbol{q}^* = \begin{bmatrix} \cos(\phi/2) \\ -\hat{\boldsymbol{e}}\,\sin(\phi/2) \end{bmatrix} \tag{1}$$

where $\boldsymbol{q}^*$ is the conjugate (the inverse rotation for a unit quaternion). A zero-length axis ($\|\boldsymbol{e}\| < 10^{-12}$) is rejected.

1.2 Quaternion (Hamilton) Product

For $\boldsymbol{p} = [p_0,\ \boldsymbol{p}_v^T]^T$ and $\boldsymbol{q} = [q_0,\ \boldsymbol{q}_v^T]^T$:

$$\boldsymbol{p} \otimes \boldsymbol{q} = \begin{bmatrix} p_0 q_0 - \boldsymbol{p}_v \cdot \boldsymbol{q}_v \\ p_0\,\boldsymbol{q}_v + q_0\,\boldsymbol{p}_v + \boldsymbol{p}_v \times \boldsymbol{q}_v \end{bmatrix} \tag{2}$$

1.3 Rotation of a Point

A point $\boldsymbol{r} \in \mathbb{R}^3$ is embedded as the pure quaternion $\boldsymbol{p} = [0,\ \boldsymbol{r}^T]^T$ and rotated by the sandwich product; the rotated point is the vector part of the result:

$$\boldsymbol{p}' = \boldsymbol{q} \otimes \boldsymbol{p} \otimes \boldsymbol{q}^* \tag{3}$$

1.4 Error Quaternion

With current attitude $\boldsymbol{q}$ and desired attitude $\boldsymbol{q}_d$ (both normalised first), the attitude error is the rotation from the desired frame to the body frame, $\boldsymbol{q}_e = \boldsymbol{q}_d^* \otimes \boldsymbol{q}$:

$$q_{e0} = q_0\,q_{d0} + \boldsymbol{q}_v \cdot \boldsymbol{q}_{dv}, \qquad \boldsymbol{q}_{ev} = q_{d0}\,\boldsymbol{q}_v - q_0\,\boldsymbol{q}_{dv} + \boldsymbol{q}_v \times \boldsymbol{q}_{dv} \tag{4}$$

Perfect alignment corresponds to $\boldsymbol{q}_e = [\pm 1, 0, 0, 0]^T$. The scalar attitude-error angle used in all plots is

$$\theta_{\text{err}} = 2\arccos(|q_{e0}|) \tag{5}$$

The absolute value accounts for the double cover ($\boldsymbol{q}$ and $-\boldsymbol{q}$ are the same attitude), so $\theta_{\text{err}} \in [0, \pi]$.

2 Secondary Attitude Representations

2.1 Direction Cosine Matrix

The direction cosine matrix (inertial $\rightarrow$ body) built from the unit quaternion is

$$C(\boldsymbol{q}) = \begin{bmatrix} q_0^2+q_1^2-q_2^2-q_3^2 & 2(q_1q_2+q_0q_3) & 2(q_1q_3-q_0q_2) \\ 2(q_1q_2-q_0q_3) & q_0^2-q_1^2+q_2^2-q_3^2 & 2(q_2q_3+q_0q_1) \\ 2(q_1q_3+q_0q_2) & 2(q_2q_3-q_0q_1) & q_0^2-q_1^2-q_2^2+q_3^2 \end{bmatrix} \tag{6}$$

2.2 Euler Angles (Yaw–Pitch–Roll, 3-2-1 Sequence)

$$\theta = -\arcsin\big(2(q_1q_3 - q_0q_2)\big) \qquad \text{(pitch, argument clipped to } [-1,1]) \tag{7}$$

$$\phi = \operatorname{atan2}\big(2(q_2q_3 + q_0q_1),\ q_0^2 - q_1^2 - q_2^2 + q_3^2\big) \qquad \text{(roll)} \tag{8}$$

$$\psi = \operatorname{atan2}\big(2(q_1q_2 + q_0q_3),\ q_0^2 + q_1^2 - q_2^2 - q_3^2\big) \qquad \text{(yaw)} \tag{9}$$

2.3 Modified Rodrigues Parameters

The MRP vector is $\boldsymbol{\sigma} = \hat{\boldsymbol{e}}\tan(\phi/4)$. To stay away from the singularity at $\phi = 360^\circ$, the shadow set is selected whenever $q_0 < 0$:

$$\boldsymbol{\sigma} = \begin{cases} \dfrac{\boldsymbol{q}_v}{1+q_0}, & q_0 \geq 0 \\ \dfrac{-\boldsymbol{q}_v}{1-q_0}, & q_0 < 0 \end{cases} \tag{10}$$

which guarantees $\|\boldsymbol{\sigma}\| \leq 1$.

3 Rotational Equations of Motion

3.1 Quaternion Kinematic Differential Equation

With body angular velocity $\boldsymbol{\omega} = [\omega_x, \omega_y, \omega_z]^T$, the kinematics $\dot{\boldsymbol{q}} = \frac{1}{2}\,\boldsymbol{q} \otimes [0,\ \boldsymbol{\omega}^T]^T$ are written in matrix form and augmented with a norm-constraint stabilisation term:

$$\dot{\boldsymbol{q}} = \frac{1}{2}\,\Omega(\boldsymbol{\omega})\,\boldsymbol{q} \ + \ c\left(1 - \boldsymbol{q}^T\boldsymbol{q}\right)\boldsymbol{q}, \qquad \Omega(\boldsymbol{\omega}) = \begin{bmatrix} 0 & -\omega_x & -\omega_y & -\omega_z \\ \omega_x & 0 & \omega_z & -\omega_y \\ \omega_y & -\omega_z & 0 & \omega_x \\ \omega_z & \omega_y & -\omega_x & 0 \end{bmatrix} \tag{11}$$

- $\frac{1}{2}\Omega(\boldsymbol{\omega})\boldsymbol{q}$ — **Kinematic Term**: Since Ω is skew-symmetric, this term alone preserves $\|\boldsymbol{q}\|$ exactly in continuous time.
- $c\,(1 - \boldsymbol{q}^T\boldsymbol{q})\,\boldsymbol{q}$ — **Constraint (Baumgarte-type) Term**: Vanishes on the unit sphere and pulls the numerical solution back towards $\|\boldsymbol{q}\| = 1$ when integration error causes drift. The gain is $c = 1$ in all simulations.

3.2 Euler's Rigid-Body Equation

For inertia matrix I and total body-frame torque $\boldsymbol{\tau}$:

$$I\dot{\boldsymbol{\omega}} = \boldsymbol{\tau} - \boldsymbol{\omega} \times (I\boldsymbol{\omega}) \quad \Longrightarrow \quad \dot{\boldsymbol{\omega}} = I^{-1}\big(\boldsymbol{\tau} - \boldsymbol{\omega} \times (I\boldsymbol{\omega})\big) \tag{12}$$

The term $\boldsymbol{\omega} \times (I\boldsymbol{\omega})$ is the gyroscopic (cross-coupling) torque. The linear solve is performed either with a pre-computed inverse I^{-1} or, optionally, with a pre-computed LU factorisation $PI = LU$ followed by forward/back substitution.

3.3 Coupled Nonlinear State-Space Form

Stacking attitude and rate gives the 7-dimensional state $\boldsymbol{x} = [\boldsymbol{q}^T,\ \boldsymbol{\omega}^T]^T \in \mathbb{R}^7$:

$$\dot{\boldsymbol{x}} = f(\boldsymbol{x}, \boldsymbol{\tau}) = \begin{bmatrix} \frac{1}{2}\,\Omega(\boldsymbol{\omega})\,\boldsymbol{q} + c\,\big(1 - \boldsymbol{q}^T\boldsymbol{q}\big)\,\boldsymbol{q} \\ I^{-1}\big(\boldsymbol{\tau} - \boldsymbol{\omega} \times (I\boldsymbol{\omega})\big) \end{bmatrix} \tag{13}$$

3.4 Conservation-Law Checks

For verifying the integrator in torque-free motion:

$$T_{\text{rot}} = \frac{1}{2}\,\boldsymbol{\omega}^T I\,\boldsymbol{\omega}, \qquad \boldsymbol{H}_B = I\boldsymbol{\omega}, \qquad \boldsymbol{H}_I = C(\boldsymbol{q})^T\,\boldsymbol{H}_B \tag{14}$$

With $\boldsymbol{\tau} = \mathbf{0}$, the rotational kinetic energy T_{rot} and the inertial angular momentum $\boldsymbol{H}_I$ must remain constant (the body-frame components $\boldsymbol{H}_B$ generally do not).

4 Numerical Integration: Classical Fourth-Order Runge–Kutta

4.1 Kinematics-Only Integrator

When $\boldsymbol{\omega}(t)$ is a prescribed function of time, only $\boldsymbol{q}$ is integrated, with step size h and $g(\boldsymbol{q}, \boldsymbol{\omega})$ denoting the right-hand side of the kinematic equation:

$$\boldsymbol{k}_1 = h\,g\big(\boldsymbol{q}_n,\ \boldsymbol{\omega}(t_n)\big) \tag{15}$$

$$\boldsymbol{k}_2 = h\,g\big(\boldsymbol{q}_n + \tfrac{1}{2}\boldsymbol{k}_1,\ \boldsymbol{\omega}(t_n + \tfrac{h}{2})\big) \tag{16}$$

$$\boldsymbol{k}_3 = h\,g\big(\boldsymbol{q}_n + \tfrac{1}{2}\boldsymbol{k}_2,\ \boldsymbol{\omega}(t_n + \tfrac{h}{2})\big) \tag{17}$$

$$\boldsymbol{k}_4 = h\,g\big(\boldsymbol{q}_n + \boldsymbol{k}_3,\ \boldsymbol{\omega}(t_n + h)\big) \tag{18}$$

$$\tilde{\boldsymbol{q}}_{n+1} = \boldsymbol{q}_n + \tfrac{1}{6}\,(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4), \qquad \boldsymbol{q}_{n+1} = \frac{\tilde{\boldsymbol{q}}_{n+1}}{\|\tilde{\boldsymbol{q}}_{n+1}\|} \tag{19}$$

4.2 Coupled-State Integrator

For the full state $\boldsymbol{x}$ with torque law $\boldsymbol{\tau}(t, \boldsymbol{x})$:

$$\boldsymbol{k}_1 = f\big(\boldsymbol{x}_n,\ \boldsymbol{\tau}(t_n, \boldsymbol{x}_n)\big) \tag{20}$$

$$\boldsymbol{k}_2 = f\big(\boldsymbol{x}_n + \tfrac{h}{2}\boldsymbol{k}_1,\ \boldsymbol{\tau}(t_n + \tfrac{h}{2},\ \boldsymbol{x}_n + \tfrac{h}{2}\boldsymbol{k}_1)\big) \tag{21}$$

$$\boldsymbol{k}_3 = f\big(\boldsymbol{x}_n + \tfrac{h}{2}\boldsymbol{k}_2,\ \boldsymbol{\tau}(t_n + \tfrac{h}{2},\ \boldsymbol{x}_n + \tfrac{h}{2}\boldsymbol{k}_2)\big) \tag{22}$$

$$\boldsymbol{k}_4 = f\big(\boldsymbol{x}_n + h\,\boldsymbol{k}_3,\ \boldsymbol{\tau}(t_n + h,\ \boldsymbol{x}_n + h\,\boldsymbol{k}_3)\big) \tag{23}$$

$$\boldsymbol{x}_{n+1} = \boldsymbol{x}_n + \frac{h}{6}\,(\boldsymbol{k}_1 + 2\boldsymbol{k}_2 + 2\boldsymbol{k}_3 + \boldsymbol{k}_4) \tag{24}$$

followed by re-normalisation of the quaternion part only, $\boldsymbol{q}_{n+1} \leftarrow \boldsymbol{q}_{n+1}/\|\boldsymbol{q}_{n+1}\|$.

Torque handling inside a step. Two modes are available:

- **Continuous feedback** (default): the torque law is re-evaluated at every RK4 stage, as written above.
- **Zero-order hold**: the torque is evaluated once, $\boldsymbol{\tau}_n = \boldsymbol{\tau}(t_n, \boldsymbol{x}_n)$, and the same value is used in all four stages. This models a sampled-data (digital) controller.

4.3 Propagation Over a Time Interval

To propagate from t_0 to t_f with $T = t_f - t_0 \geq 0$ and tolerance $\varepsilon = 10^{-9}$:

1. Number of full steps: $n_{\text{full}} = \lfloor T/h + \varepsilon \rfloor$.
2. Remainder: $r = T - n_{\text{full}}\, h$. A final partial step of size r is taken only if $r > \varepsilon h$.
3. Take n_{full} steps of size h. After each step the time is reset to $t_i = t_0 + i\, h$ (rather than accumulated as $t_{i-1} + h$) to avoid floating-point round-off drift.
4. If required, take the partial step and set the final time exactly to t_f.
5. Return the time history and the $(n+1) \times 7$ state history, including the initial state.

5 Attitude Regulation: Quaternion-Feedback PD Law

5.1 Control Law

For regulation to a fixed target attitude $\boldsymbol{q}_d$:

$$\boldsymbol{\tau} = -K_p\, \operatorname{sgn}(q_{e0})\, \boldsymbol{q}_{ev} \;-\; K_d\, \boldsymbol{\omega}, \qquad \operatorname{sgn}(q_{e0}) = \begin{cases} +1, & q_{e0} \geq 0 \\ -1, & q_{e0} < 0 \end{cases} \tag{25}$$

- $-K_p\, \operatorname{sgn}(q_{e0})\, \boldsymbol{q}_{ev}$ — **Proportional (Attitude) Term**: Restoring torque along the error rotation axis. Since $\boldsymbol{q}_{ev} = \hat{\boldsymbol{e}} \sin(\theta_{\text{err}}/2)$, it behaves as a nonlinear spring.
- $\operatorname{sgn}(q_{e0})$ — **Shortest-Path Selection**: $\boldsymbol{q}_e$ and $-\boldsymbol{q}_e$ represent the same attitude. The sign factor always drives the body through the rotation of less than 180° and prevents unwinding.
- $-K_d\, \boldsymbol{\omega}$ — **Derivative (Rate Damping) Term**: Dissipates rotational kinetic energy. K_d may be a scalar or a 3×3 matrix.

5.2 Lyapunov Stability Analysis

The candidate function is the rotational kinetic energy plus an attitude "potential":

$$V(\boldsymbol{q}_e, \boldsymbol{\omega}) = \frac{1}{2}\, \boldsymbol{\omega}^T I\, \boldsymbol{\omega} + 2K_p\, (1 - |q_{e0}|) \tag{26}$$

$V \geq 0$, with $V = 0$ only at $\boldsymbol{\omega} = \mathbf{0}$ and $q_{e0} = \pm 1$. For a constant target the error kinematics give $\dot{q}_{e0} = -\frac{1}{2}\, \boldsymbol{q}_{ev} \cdot \boldsymbol{\omega}$, hence

$$\begin{aligned} \dot{V} &= \boldsymbol{\omega}^T \big(\boldsymbol{\tau} - \boldsymbol{\omega} \times I\boldsymbol{\omega}\big) + K_p\, \operatorname{sgn}(q_{e0})\, \boldsymbol{q}_{ev} \cdot \boldsymbol{\omega} \\ &= \boldsymbol{\omega}^T \big(\boldsymbol{\tau} + K_p\, \operatorname{sgn}(q_{e0})\, \boldsymbol{q}_{ev}\big) \qquad (\text{since } \boldsymbol{\omega} \cdot (\boldsymbol{\omega} \times I\boldsymbol{\omega}) = 0) \end{aligned} \tag{27}$$

Substituting the control law:

$$\dot{V} = -\boldsymbol{\omega}^T K_d\, \boldsymbol{\omega} \;\leq\; 0 \quad \text{for } K_d > 0 \tag{28}$$

$\dot{V}$ is only negative *semi*-definite (it vanishes whenever $\boldsymbol{\omega} = \mathbf{0}$), so asymptotic convergence follows from LaSalle's invariance principle: $\boldsymbol{\omega} \equiv \mathbf{0}$ requires $\dot{\boldsymbol{\omega}} = \mathbf{0}$, which forces $\boldsymbol{q}_{ev} = \mathbf{0}$, i.e. the target attitude. Note that the gyroscopic term does no work and therefore need not be cancelled for regulation.

5.3 Gain Selection from Settling Time and Damping Ratio

For small errors $\boldsymbol{q}_{ev} \approx \frac{1}{2}\boldsymbol{\theta}$, so each principal axis behaves as the second-order system

$$I_i\ddot{\theta}_i + K_d\,\dot{\theta}_i + \frac{K_p}{2}\,\theta_i = 0 \quad \Longrightarrow \quad \omega_n^2 = \frac{K_p}{2I_i}, \qquad 2\zeta\omega_n = \frac{K_d}{I_i} \tag{29}$$

Using the 2 % settling-time approximation $t_s \approx 4/(\zeta\omega_n)$, and sizing on the largest principal inertia $I_{\text{ref}} = \max_i I_{ii}$:

$$\omega_n = \frac{4}{\zeta\, t_s}, \qquad K_p = 2\, I_{\text{ref}}\,\omega_n^2 \tag{30}$$

$$K_d = \begin{cases} 2\zeta\omega_n I_{\text{ref}} \cdot \mathbf{1}_{3\times3}, & \text{isotropic option} \\ \operatorname{diag}\!\left(2\zeta\omega_n\sqrt{I_{\text{ref}}\, I_{ii}}\right), & \text{per-axis option} \end{cases} \tag{31}$$

where $\mathbf{1}_{3\times3}$ is the identity matrix. In the per-axis option the scalar K_p gives axis i a natural frequency $\omega_n\sqrt{I_{\text{ref}}/I_{ii}}$, and the $\sqrt{I_{\text{ref}}I_{ii}}$ scaling of K_d makes the damping ratio equal to ζ on every axis.

6 Attitude Tracking: Feedforward plus PD

6.1 Rate Error

The reference supplies $\boldsymbol{q}_d(t)$, $\boldsymbol{\omega}_d(t)$ and $\dot{\boldsymbol{\omega}}_d(t)$, the latter two expressed in the desired frame. They are mapped into the body frame with the DCM of the error quaternion, $C_e = C(\boldsymbol{q}_e)$:

$$\boldsymbol{\omega}_d^B = C_e\,\boldsymbol{\omega}_d, \qquad \boldsymbol{\omega}_e = \boldsymbol{\omega} - \boldsymbol{\omega}_d^B \tag{32}$$

6.2 Control Law

$$\boldsymbol{\tau} = \underbrace{\boldsymbol{\omega}\times(I\boldsymbol{\omega})}_{\text{gyroscopic cancellation}} + \underbrace{I\left(C_e\,\dot{\boldsymbol{\omega}}_d - \boldsymbol{\omega}_e\times\boldsymbol{\omega}_d^B\right)}_{\text{reference acceleration feedforward}} - \underbrace{K_p\,\operatorname{sgn}(q_{e0})\,\boldsymbol{q}_{ev}}_{\text{attitude feedback}} - \underbrace{K_d\,\boldsymbol{\omega}_e}_{\text{rate-error damping}} \tag{33}$$

The feedforward bracket is the body-frame time derivative of $\boldsymbol{\omega}_d^B$, because $\dot{C}_e = -[\boldsymbol{\omega}_e\times]\,C_e$:

$$\frac{d}{dt}\left(C_e\,\boldsymbol{\omega}_d\right) = C_e\,\dot{\boldsymbol{\omega}}_d - \boldsymbol{\omega}_e\times\boldsymbol{\omega}_d^B \tag{34}$$

Substituting into Euler's equation gives the closed-loop error dynamics

$$I\,\dot{\boldsymbol{\omega}}_e = -K_p\,\operatorname{sgn}(q_{e0})\,\boldsymbol{q}_{ev} - K_d\,\boldsymbol{\omega}_e \tag{35}$$

which has the same structure as the regulation problem, so the tracking error converges with the same Lyapunov argument (with $\boldsymbol{\omega}$ replaced by $\boldsymbol{\omega}_e$). With a perfect model and matched initial conditions the tracking error is identically zero.

7 Smooth Reference Generation: Quintic Polynomial Ramp

A step change in the commanded attitude makes $\boldsymbol{\omega}_d$ and $\dot{\boldsymbol{\omega}}_d$ impulsive and produces torque spikes. Instead, a rotation of total angle A about a fixed unit axis $\hat{\boldsymbol{e}}$ is commanded between t_s and $t_s + T_r$ using a C^2-smooth profile. With normalised time $\sigma = (t - t_s)/T_r$:

$$s(\sigma) = 10\sigma^3 - 15\sigma^4 + 6\sigma^5 \tag{36}$$

$$\dot{s} = \frac{30\sigma^2 - 60\sigma^3 + 30\sigma^4}{T_r} \tag{37}$$

$$\ddot{s} = \frac{60\sigma - 180\sigma^2 + 120\sigma^3}{T_r^2} \tag{38}$$

with $(s, \dot{s}, \ddot{s}) = (0, 0, 0)$ for $t \le t_s$ and $(1, 0, 0)$ for $t \ge t_s + T_r$. This is the unique fifth-order polynomial satisfying the six boundary conditions

$$s(0) = 0, \quad s(1) = 1, \quad \dot{s}(0) = \dot{s}(1) = 0, \quad \ddot{s}(0) = \ddot{s}(1) = 0 \tag{39}$$

so angle, rate and acceleration are all continuous. The reference signals are

$$\boldsymbol{q}_d(t) = \begin{bmatrix} \cos\!\left(\frac{1}{2} A\, s\right) \\ \hat{\boldsymbol{e}}\, \sin\!\left(\frac{1}{2} A\, s\right) \end{bmatrix}, \qquad \boldsymbol{\omega}_d(t) = \hat{\boldsymbol{e}}\, A\, \dot{s}, \qquad \dot{\boldsymbol{\omega}}_d(t) = \hat{\boldsymbol{e}}\, A\, \ddot{s} \tag{40}$$

For a fixed-axis rotation the components of $\boldsymbol{\omega}_d$ are identical in the inertial and desired frames. The peak commanded rate is $|\boldsymbol{\omega}_d|_{\max} = 1.875\, A/T_r$, reached at $\sigma = \frac{1}{2}$.

8 Integral Action and Anti-Windup (PID)

8.1 Integral State

A PD law cannot reject a constant disturbance torque: at rest, $\boldsymbol{\tau}_{\text{dist}} = K_p\, \text{sgn}(q_{e0})\, \boldsymbol{q}_{ev}$ leaves a steady-state attitude error. An integral state $\boldsymbol{z} \in \mathbb{R}^3$ on the vector part of the error quaternion is therefore added:

$$\dot{\boldsymbol{z}} = \text{sgn}(q_{e0})\, \boldsymbol{q}_{ev} \tag{41}$$

discretised with forward Euler at the control sample time Δt, including both anti-windup mechanisms:

$$\boldsymbol{z}_{k+1} = \begin{cases} \text{clip}\big(\boldsymbol{z}_k + \text{sgn}(q_{e0})\, \boldsymbol{q}_{ev}\, \Delta t,\ -z_{\max},\ +z_{\max}\big), & \text{actuator not saturated} \\ \boldsymbol{z}_k, & \text{actuator saturated} \end{cases} \tag{42}$$

- **Clamping**: Each component of $\boldsymbol{z}$ is limited to $\pm z_{\max}$, bounding the integral torque to $K_i z_{\max}$.
- **Conditional Integration**: The integrator is frozen while the actuator reports saturation, so it does not accumulate error that the actuator cannot act upon.

8.2 PID Control Laws

The integral term is subtracted from the PD laws of the previous sections (K_p, K_i may be per-axis vectors applied element-wise, denoted $\odot$):

$$\text{Regulation:} \quad \boldsymbol{\tau} = -K_p \odot \text{sgn}(q_{e0})\, \boldsymbol{q}_{ev} - K_d\, \boldsymbol{\omega} - K_i \odot \boldsymbol{z} \tag{43}$$

$$\text{Tracking:} \quad \boldsymbol{\tau} = \boldsymbol{\omega} \times (I\boldsymbol{\omega}) + I\left(C_e\, \dot{\boldsymbol{\omega}}_d - \boldsymbol{\omega}_e \times \boldsymbol{\omega}_d^B\right) - K_p \odot \text{sgn}(q_{e0})\, \boldsymbol{q}_{ev} - K_d\, \boldsymbol{\omega}_e - K_i \odot \boldsymbol{z} \tag{44}$$

Because the controller now has internal state, it is evaluated exactly once per control sample and *not* inside the RK4 stages.

8.3 Stability Limit on the Integral Gain

Linearising one axis ($\boldsymbol{q}_{ev} \approx \frac{1}{2}\theta$) gives the characteristic polynomial

$$s^3 + \frac{K_d}{I}\, s^2 + \frac{K_p}{2I}\, s + \frac{K_i}{2I} = 0 \tag{45}$$

and the Routh–Hurwitz criterion ($a_2 a_1 > a_0$) yields the upper bound

$$0 < K_i < \frac{K_p\, K_d}{I} \tag{46}$$

(neglecting actuator lag, which reduces the real margin further).

9 Actuator Model

9.1 First-Order Lag

The actuator is modelled as a first-order system with time constant T_a:

$$T_a\,\dot{\boldsymbol{\tau}}_{\text{act}} + \boldsymbol{\tau}_{\text{act}} = \boldsymbol{\tau}_{\text{cmd}} \tag{47}$$

Discretising with backward (implicit) Euler, which is unconditionally stable for any Δt:

$$\boldsymbol{\tau}_{\text{act},k+1} = (1-\alpha_a)\,\boldsymbol{\tau}_{\text{act},k} + \alpha_a\,\boldsymbol{\tau}_{\text{cmd},k}, \qquad \alpha_a = \frac{\Delta t}{T_a + \Delta t} \tag{48}$$

9.2 Magnitude Saturation

The limit is applied to the torque vector norm, so the direction of the torque is preserved:

$$\boldsymbol{\tau}_{\text{act}} \leftarrow \begin{cases} \boldsymbol{\tau}_{\text{act}}, & \|\boldsymbol{\tau}_{\text{act}}\| \le \tau_{\max} \\ \tau_{\max}\,\dfrac{\boldsymbol{\tau}_{\text{act}}}{\|\boldsymbol{\tau}_{\text{act}}\|}, & \|\boldsymbol{\tau}_{\text{act}}\| > \tau_{\max} \end{cases} \tag{49}$$

The flag saturated $= \big(\|\boldsymbol{\tau}_{\text{act}}\| > \tau_{\max}\big)$ is passed back to the PID controller on the next sample for conditional integration.

9.3 Torque Limit from Fin Authority

For the missile, the available control torque is the aerodynamic moment of a fully deflected fin:

$$\tau_{\max} = \bar{q}\,S\,\bar{c}\,|C_{m\delta}|\,\delta_{\max} \tag{50}$$

10 Missile Model: Atmosphere, Inertia and Aerodynamic Moment

10.1 Standard Atmosphere

With $g_0 = 9.80665$ m/s^2, $R = 287.05$ J/(kg K), $\gamma = 1.4$ and lapse rate $\lambda = 0.0065$ K/m, at altitude h:

$$T = \begin{cases} 288.15 - \lambda h, & h \le 11\,000 \text{ m} \\ 216.65 \text{ K}, & h > 11\,000 \text{ m} \end{cases} \tag{51}$$

$$p = \begin{cases} 101\,325\left(\dfrac{T}{288.15}\right)^{g_0/(\lambda R)}, & h \le 11\,000 \text{ m} \\ 22\,632.06\,\exp\!\left(-\dfrac{g_0\,(h-11\,000)}{R\,T}\right), & h > 11\,000 \text{ m} \end{cases} \tag{52}$$

- **Air Density** (ideal gas law): $\rho = \dfrac{p}{R\,T}$
- **Speed of Sound**: $a = \sqrt{\gamma R T}$
- **Flight Velocity**: $V = Ma$
- **Dynamic Pressure**: $\bar{q} = \frac{1}{2}\rho V^2$

10.2 Inertia Matrix

The missile is treated as a uniform solid cylinder of mass m, radius r and length L:

$$I_{yy} = I_{zz} = \frac{1}{12}mL^2 + \frac{1}{4}mr^2, \qquad I_{xx} = \frac{1}{2}mr^2, \qquad I = \operatorname{diag}(I_{xx},\ I_{yy},\ I_{zz}) \tag{53}$$

10.3 Angle of Attack

For the rotation-only demonstration the velocity vector is assumed fixed along the inertial x-axis, so the angle of attack equals the pitch angle of the yaw–pitch–roll sequence:

$$\alpha = \theta = \arcsin\bigl(2(q_0 q_2 - q_1 q_3)\bigr) \tag{54}$$

10.4 Pitch-Channel Aerodynamic Moment

$$C_m = C_{m\alpha}\,\alpha + C_{m\delta}\,\delta, \qquad M_y = \bar{q}\,S\,\bar{c}\,C_m, \qquad \boldsymbol{M}_{\text{aero}}(\boldsymbol{q}) = \begin{bmatrix} 0 \\ M_y \\ 0 \end{bmatrix} \tag{55}$$

With $C_{m\alpha} < 0$ the moment is statically stabilising: it opposes any pitch away from $\alpha = 0$ and therefore acts as a disturbance that grows with the commanded pitch angle. The fin deflection δ is set to zero inside this term, because the control moment is supplied through the actuator torque $\boldsymbol{\tau}_{\text{act}}$.

10.5 Imperfect Model for Feedforward

The controller holds its own copy of the aero model with every coefficient multiplied by a scale factor κ:

$$\hat{C}_{m\alpha} = \kappa\,C_{m\alpha}, \qquad \hat{C}_{m\delta} = \kappa\,C_{m\delta}, \qquad \hat{\boldsymbol{M}}_{\text{aero}} = \kappa\,\boldsymbol{M}_{\text{aero}} \tag{56}$$

With $\kappa = 0.8$ the controller under-estimates the moment by 20 %. After cancellation, the residual disturbance $(1-\kappa)\,\boldsymbol{M}_{\text{aero}}$ remains, and it is this residual that the integral term must remove.

11 Closed-Loop Missile Pitch Simulation

11.1 Per-Axis Gain Scaling

The roll inertia is roughly 90 times smaller than the pitch/yaw inertia. A single scalar K_p sized on $I_{\max}$ would make the roll loop faster than the actuator bandwidth and unstable. The reference gains are therefore scaled per axis with $s_i = I_{ii}/I_{\max}$:

$$K_{p,i} = s_i\,K_p^{\text{ref}} = 2\,I_{ii}\,\omega_n^2, \qquad K_{d,i} = s_i\,K_d^{\text{ref}} = 2\zeta\omega_n I_{ii}, \qquad K_{i,i} = s_i\,K_I \tag{57}$$

so every axis has the same natural frequency ω_n and damping ratio ζ.

11.2 Commanded Torque

$$\boldsymbol{\tau}_{\text{cmd}} = \boldsymbol{\tau}_{\text{PID}} - \hat{\boldsymbol{M}}_{\text{aero}}(\boldsymbol{q}) \tag{58}$$

where $\boldsymbol{\tau}_{\text{PID}}$ is the tracking PID law and the second term cancels the (imperfectly known) aerodynamic moment.

11.3 Total Torque on the Plant

$$\boldsymbol{\tau}(t,\boldsymbol{x}) = \underbrace{\boldsymbol{\tau}_{\text{act},k}}_{\text{held over the sample}} + \underbrace{\boldsymbol{M}_{\text{aero}}\bigl(\boldsymbol{q}(t)\bigr)}_{\text{re-evaluated in each RK4 stage}} \tag{59}$$

The actuator output is a zero-order hold over the control sample, whereas the true aerodynamic moment depends on attitude and is continuous, so it is evaluated inside the RK4 stages.

11.4 Simulation Algorithm

For $k = 0, 1, \ldots, N$ with $N = \text{round}(t_f/\Delta t)$ and $t_k = k\,\Delta t$:

1. Read the plant state $\boldsymbol{q}_k$, $\boldsymbol{\omega}_k$.
2. Evaluate the quintic reference: $\boldsymbol{q}_d(t_k)$, $\boldsymbol{\omega}_d(t_k)$, $\dot{\boldsymbol{\omega}}_d(t_k)$.
3. Compute the error quaternion $\boldsymbol{q}_e = \boldsymbol{q}_d^* \otimes \boldsymbol{q}_k$.
4. Update the integral state (frozen if the actuator saturated on the previous sample) and compute $\boldsymbol{\tau}_{\text{PID}}$.
5. Subtract the modelled aero moment: $\boldsymbol{\tau}_{\text{cmd}} = \boldsymbol{\tau}_{\text{PID}} - \hat{\boldsymbol{M}}_{\text{aero}}(\boldsymbol{q}_k)$.
6. Pass $\boldsymbol{\tau}_{\text{cmd}}$ through the actuator lag and saturation to obtain $\boldsymbol{\tau}_{\text{act},k}$.
7. Log θ_{err}, θ, θ_d, $\boldsymbol{\tau}_{\text{cmd}}$, $\boldsymbol{\tau}_{\text{act}}$, $\boldsymbol{M}_{\text{aero}}$ and $\boldsymbol{z}$.
8. If $k < N$: advance the plant one RK4 step of size Δt with the total torque of the previous subsection.

The simulation is run twice, with $K_I > 0$ (PID) and with $K_I = 0$ (PD), to isolate the effect of integral action on the steady-state error.

12 Simulation Scenarios and Parameters

12.1 Scenario 1: Rest-to-Rest Regulation

A rigid body is regulated from a 70° initial rotation about the body x-axis to the identity attitude with the PD law (continuous feedback). The Lyapunov function $V(t)$ is evaluated along the trajectory and plotted on a logarithmic scale to confirm monotonic decay.

12.2 Scenario 2: Smooth Trajectory Tracking

The same rigid body tracks a 40° quintic ramp about the z-axis using the feedforward-plus-PD tracking law, starting from rest at the identity attitude.

Parameter	**Scenario 1**	**Scenario 2**
Inertia I [kg m^2]	diag(10, 15, 20)	diag(10, 15, 20)
Initial attitude	70° about x	identity
Initial rate $\boldsymbol{\omega}_0$	$\mathbf{0}$	$\mathbf{0}$
Target	identity	40° about z, $t_s = 2$ s, $T_r = 6$ s
Settling time t_s / damping ζ	6 s / 0.8	5 s / 0.85
$\omega_n = 4/(\zeta t_s)$ [rad/s]	0.833	0.941
$K_p = 2I_{\text{ref}}\,\omega_n^2$	27.78	35.43
$K_d = 2\zeta\omega_n I_{\text{ref}}$ (isotropic)	26.67	32.00
Step size h / final time	0.01 s / 15 s	0.01 s / 15 s

12.3 Scenario 3: Missile Pitch Tracking with Aero Moment, Actuator and Integral Action

Parameter	Value
Altitude h / Mach number M	10 000 m / 2.5
Mass m, radius r, length L	180 kg, 0.113 m, 2.65 m
Reference area S, chord $\bar{c}$	0.04 m^2, 0.225 m
$C_{Z\alpha}$, $C_{m\alpha}$, $C_{m\delta}$, $C_{Z\delta}$ [per rad]	-12, -15, -20, -2
Pitch command	8° about y, $t_s = 1$ s, $T_r = 3$ s
Settling time / damping ratio	1 s / 0.8
Integral gain K_I / integral limit $z_{\max}$	8000 / 0.5
Actuator time constant T_a / fin limit $\delta_{\max}$	0.08 s / 15°
Aero model scale κ	0.8
Sample time Δt / final time	0.01 s / 12 s

Quantities that follow from these parameters through the formulas above:

Derived quantity	Value
Temperature T	223.15 K
Pressure p	26 436 Pa
Density ρ	0.4127 kg/m^3
Speed of sound a / velocity V	299.46 m/s / 748.65 m/s
Dynamic pressure $\bar{q}$	115 657 Pa
$I_{yy} = I_{zz}$ / I_{xx}	105.91 / 1.149 kg m^2 (ratio ≈ 92)
ω_n	5.0 rad/s
K_p^{ref} / K_d^{ref}	5295.6 / 847.3
Integral stability limit $K_p K_d / I_{yy}$	42 365 (K_I is $\approx 19\,\%$ of it)
Actuator torque limit $\tau_{\max}$	5450 N m
Aero moment at $\alpha = 8^\circ$	-2180 N m

13 Symbol Definitions

- $\boldsymbol{q}$, $\boldsymbol{q}_d$, $\boldsymbol{q}_e$ **(Quaternions):** Current, desired and error attitude (scalar-first, unit norm).
- $\boldsymbol{\omega}$, $\boldsymbol{\omega}_d$, $\boldsymbol{\omega}_e$ **(Angular Velocities):** Body rate, desired rate (desired frame) and rate error (body frame), in rad/s.
- I **(Inertia Matrix):** Body-frame moment of inertia tensor, kg m^2.
- $\boldsymbol{\tau}$, $\boldsymbol{\tau}_{\mathrm{cmd}}$, $\boldsymbol{\tau}_{\mathrm{act}}$ **(Torques):** Total, commanded and actuator-delivered body torque, N m.
- K_p, K_d, K_i **(Controller Gains):** Proportional, derivative and integral gains.
- $\boldsymbol{z}$ **(Integral State):** Accumulated vector part of the error quaternion.
- c **(Constraint Gain):** Quaternion norm stabilisation gain.
- ζ, ω_n, t_s **(Loop Specifications):** Damping ratio, natural frequency and 2 % settling time.
- T_a, $\tau_{\max}$ **(Actuator Parameters):** Time constant and torque magnitude limit.
- C_e **(Error DCM):** Direction cosine matrix from the desired frame to the body frame.
- κ **(Model Scale):** Ratio of the controller's aero coefficients to the true ones.